

3.5 Markov Property

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Introduction

- In this section, we show that Brownian motion is a Markov process and discuss its *transition density*.

Theorem 3.5.1

- Let $W(t)$, $t \geq 0$, be a Brownian motion and let $F(t)$, $t \geq 0$, be a filtration for this Brownian motion (see Definition 3.3.3). Then $W(t)$, $t \geq 0$, is a Markov process.

Theorem 3.5.1 Proof (1)

- According to Definition 2.3.6 (Markov process), we must show that
 - whenever $0 \leq s \leq t$ and
 - f is a Borel-measurable function,there is another Borel-measurable function g such that

$$E[f(W(t))|F(s)] = g(W(s)). \quad (3.5.1)$$

- $E[f(W(t))|F(s)] = E[f((W(t)-W(s))+W(s))|F(s)]$.
 - The random variable $W(t)-W(s)$ is independent of $F(s)$
 - The random variable $W(s)$ is $F(s)$ -measurable
 - This permits us to apply the Independence Lemma, Lemma 2.3.4.

Theorem 3.5.1 Proof (2)

- Compute $E[f((W(t)-W(s))+W(s))|F(s)]$
 - Replace $W(s)$ by a dummy variable x to hold it constant
 - Define $g(x) = E f(W(t)-W(s)+x)$
 - $W(t)-W(s)$ is normally distributed with mean=0 and variance= $t-s$
 - Therefore,

$$g(x) = \frac{1}{\sqrt{2\pi(t-s)}} \int_{-\infty}^{\infty} f(\omega + x) e^{-\frac{\omega^2}{2(t-s)}} d\omega$$

- Replace $x=W(s)$, then the equation $E[f(W(t))|F(s)]=g(W(s))$ holds.

$$g(x) = \frac{1}{\sqrt{2\pi(t-s)}} \int_{-\infty}^{\infty} f(\omega + x) e^{-\frac{\omega^2}{2(t-s)}} d\omega$$

- Let $\tau = t-s$, $y = \omega + x$, then

$$g(x) = \frac{1}{\sqrt{2\pi\tau}} \int_{-\infty}^{\infty} f(y) e^{-\frac{(y-x)^2}{2\tau}} dy$$

- We define the *transition density* $p(\tau, x, y)$ for Brownian motion to be

$$p(\tau, x, y) = \frac{1}{\sqrt{2\pi\tau}} e^{-\frac{(y-x)^2}{2\tau}}$$

- So that we can rewrite $g(x)$ as

$$g(x) = \int_{-\infty}^{\infty} f(y) p(\tau, x, y) dy$$

- And $E[f(W(t))|F(s)] = g(W(s))$ as

$$E[f(W(t))|F(s)] = \int_{-\infty}^{\infty} f(y) p(\tau, W(s), y) dy$$

$$E[f(W(t))|F(s)] = \int_{-\infty}^{\infty} f(y)p(\tau, W(s), y)dy$$

- Conditioned on the information in $F(s)$
 - (which contains all the information obtained by observing the Brownian motion up to and including time s)
- The conditional density of $W(t)$ is $p(\tau, W(s), y)$
 - This is a density in the variable y
 - This density is normal with mean $W(s)$, variance $\tau = t-s$
- In particular, the only information from $F(s)$ that is relevant is the value of $W(s)$
- The fact that only $W(s)$ is relevant is the essence of the Markov property.